PART 2: LOGIC SIMPLIFICATION

BOOLEAN FUNCTIONS AND FORMS

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BASIC THEOREMS OF BOOLEAN ALGEBRA

Postulates and Theorems of Boolean Algebra

Postulate 2	(a)	$x + 0 = x$	(b)	$x \cdot 1 = x$
Postulate 5	(a)	$x + x' = 1$	(b)	$x \cdot x' = 0$
Theorem 1	(a)	$x + x = x$	(b)	$x \cdot x = x$
Theorem 2	(a)	$x + 1 = 1$	(b)	$x \cdot 0 = 0$
Theorem 3, involution		$(x')' = x$		
Postulate 3, commutative	(a)	$x + y = y + x$	(b)	$xy = yx$
Theorem 4, associative	(a)	$x + (y + z) = (x + y) + z$	(b)	$x(yz) = (xy)z$
Postulate 4, distributive	(a)	$x(y + z) = xy + xz$	(b)	$x + yz = (x + y)(x + z)$
Theorem 5, DeMorgan	(a)	$(x + y)' = x'y'$	(b)	$(xy)' = x' + y'$
Theorem 6, absorption	(a)	$x + xy = x$	(b)	$x(x + y) = x$

BOOLEAN FUNCTIONS

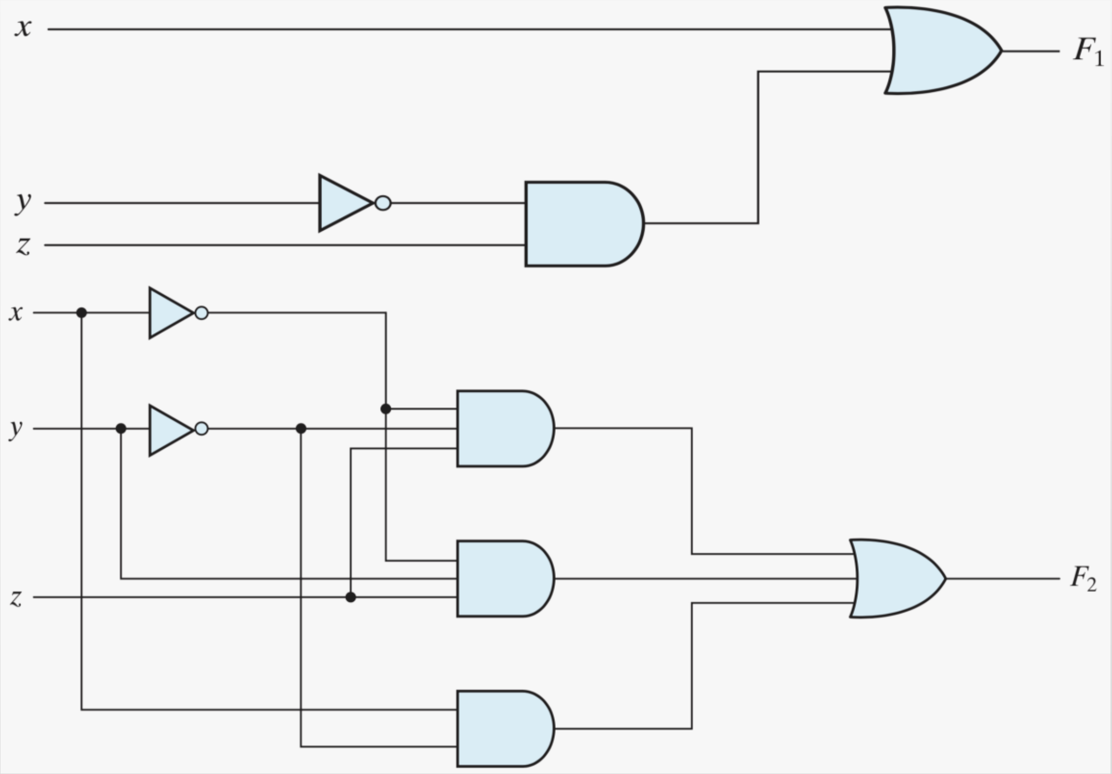
- Boolean Function described by an algebraic expression consists of binary variables, the constants 0 and 1, and logic operation symbols.
- A Boolean function can be represented in a truth table.
- The number of rows in the truth table is 2^n, where n is the number of variables in the function.
  - Example: $F_1 = x + y'z$
  - Example: $F_2 = x'y'z + x'yz + xy'$

x	y	z	F_1	F_2
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	1
1	1	0	1	0
1	1	1	1	0

BOOLEAN TO GATE IMPLEMENTATION

- Example: $F_1 = x + y'z$

- Example: $F_2 = x'y'z + x'yz + xy'$



BOOLEAN LOGIC SIMPLIFICATION

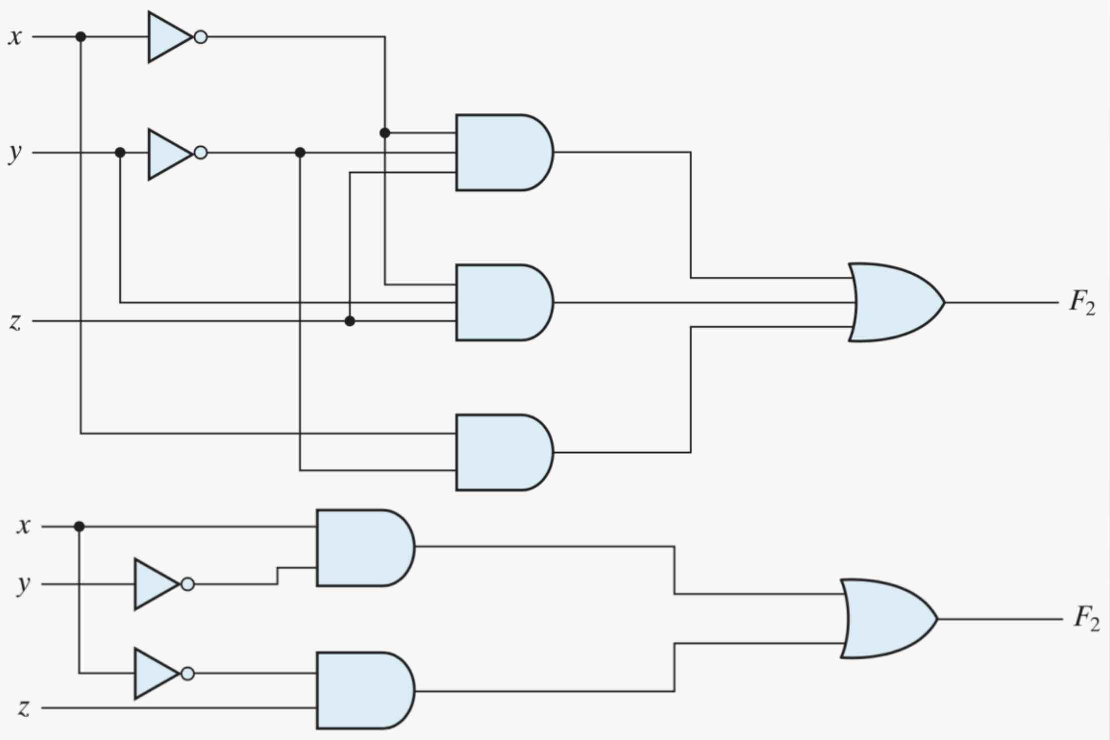
- There is only one way that a Boolean function can be represented in a truth table. However, when the function is in algebraic form, it can be expressed in a variety of ways, all of which have equivalent logic.
- The particular expression used to represent the function will dictate the interconnection of gates in the logic-circuit diagram.
- Using the rules of Boolean algebra to obtain a simpler expression for the same function can help reduce the number of gates in the circuit and the number of inputs to the gates.

BOOLEAN LOGIC SIMPLIFICATION

- Example: $F_2 = x'y'z + x'yz + xy'$

$$= x'z(y'+y) + xy'$$

$$= x'z + xy'$$



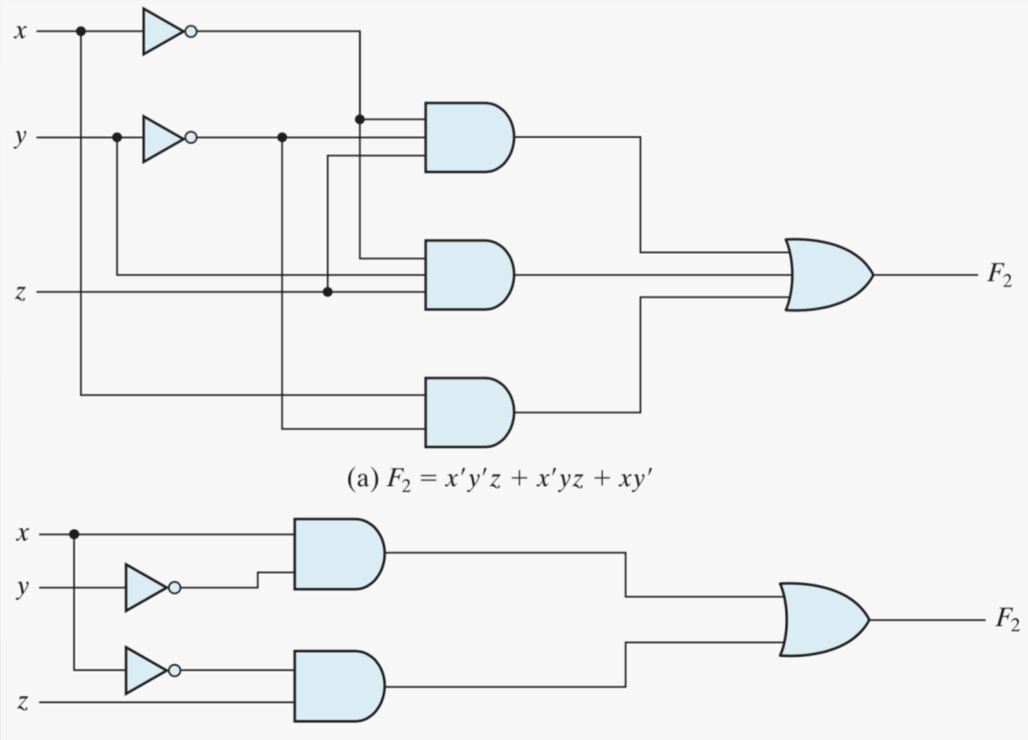
(b) $F_2 = xy' + x'z$

LITERALS AND TERMS

- A literal is a single variable within a term, in complemented or un-complemented form.
- In Boolean expression, each term requires a gate.
  - Example: How many terms and literals are in (a) & (b)?

a) 8 literals and 3 terms

b) 4 literals and 2 terms



(a) $F_2 = x'y'z + x'yz + xy'$

(b) $F_2 = xy' + x'z$

COMPLEMENT OF A FUNCTION

- The complement of function F can be derived algebraically through DeMorgan's theorems.

$$(A + B + C + D + \cdots + F)' = A'B'C'D' \dots F'$$
$$(ABCD \dots F)' = A' + B' + C' + D' + \cdots + F'$$

  - Example: Take the complement of following function: $F_2 = x(y'z' + yz)$
  - Trick: Take the dual of function and complement each literal.
  - Solution:

$$\begin{aligned} F_2' = [x(y'z' + yz)]' &= x' + (y'z' + yz)' = x' + (y'z')'(yz)' \\ &= x' + (y + z)(y' + z') \\ &= x' + yz' + y'z \end{aligned}$$

ACTIVITY 01

- Find the complement of following function: $F = x'yz' + x'y'z$

CANONICAL OR STANDARD FORMS

- There are many ways of writing Boolean Algebra for logic implementation.
- The three types are Canonical Form, Standard Form and Nonstandard Form.
- For Canonical Form, we first need to understand what minterms and maxterms are.

CANONICAL FORMS

- Boolean functions expressed as a sum of minterms or product of maxterms are said to be in Canonical Form.

Minterms and Maxterms for Three Binary Variables

			Minterms		Maxterms	
x	**y**	**z**	**Term**	**Designation**	**Term**	**Designation**
0	0	0	$x'y'z'$	m_0	$x + y + z$	M_0
0	0	1	$x'y'z$	m_1	$x + y + z'$	M_1
0	1	0	$x'yz'$	m_2	$x + y' + z$	M_2
0	1	1	$x'yz$	m_3	$x + y' + z'$	M_3
1	0	0	$xy'z'$	m_4	$x' + y + z$	M_4
1	0	1	$xy'z$	m_5	$x' + y + z'$	M_5
1	1	0	xyz'	m_6	$x' + y' + z$	M_6
1	1	1	xyz	m_7	$x' + y' + z'$	M_7

CANONICAL FORMS

n variables forming an <u>AND</u> term with each variable being primed or unprimed, provide 2^n possible combinations called Minterms or Standard Products.

- Each minterm is obtained from an AND term of the n variables, with each variable being primed if the corresponding bit of the binary number is a 0 and unprimed if a 1.

n variables forming an <u>OR</u> term with each variable being primed or unprimed, provide 2^n possible combinations called Maxterms or Standard Sums.

- Each maxterm is obtained from an OR term of the n variables, with each variable being unprimed if the corresponding bit of the binary number is a 0 and primed if a 1.

SUM OF MINTERMS

A Boolean function can be expressed algebraically from a given truth table by forming a minterm for each combination of the variables that produces a 1 in the function and then taking the OR of all those terms.

- Example: Write a Boolean function of f_1 and f_2 from following truth table.

$$f_1 = x'y'z + xy'z' + xyz = m_1 + m_4 + m_7$$

$$f_2 = x'yz + xy'z + xyz' + xyz = m_3 + m_5 + m_6 + m_7$$

x	y	z	Function f_1	Function f_2
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

ACTIVITY 01

Boolean Function Complement

- Take the complement of the following function:

$$f_1 = x'y'z + xy'z' + xyz = m_1 + m_4 + m_7$$

- $f'_1 = x'y'z' + x'yz' + xy'z' + xy'z + xyz'$
- Complementing above equation:
- $f_1 = (x+y+z)(x+y'+z)(x+y'+z')(x'+y+z')(x'+y'+z)$
- $f_1 = M0.M2.M3.M5.M6$

x	y	z	Function f_1
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

PRODUCT OF MAXTERMS

Any Boolean function can be expressed as a product of maxterms (with "product" meaning the ANDing of terms).

- Each maxterm is complement of its corresponding minterm and vice versa.

The maxterms whose product defines the Boolean function are those which give the 0's of the function in a truth table.

CANONICAL FORMS SUMMARY

Sum of Minterms

1. Expand the expression into a sum of AND terms
2. Each term is then inspected to see if it contains all the variables
3. If it misses one or more variables, it is ANDed with an expression such as $x + x'$, where x is one of the missing variables

- Alternatively, the minterms whose sum defines the Boolean function are those which give the 1's of the function in a truth table.

Product of Maxterms

1. Bring the expression into a form of OR terms
2. Use the distributive law $A + BC = (A + B)(A + C)$
3. Then any missing variable x in each OR term is ORed with xx'.

- Alternatively, the maxterms whose product defines the Boolean function are those which give the 0's of the function in a truth table.

ACTIVITY 02

Sum of minterms and product of maxterms

- $F = A + B'C$

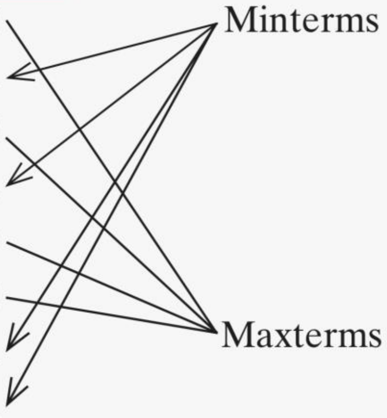
1. Express the above Boolean function as a sum of minterms.
2. Express the above Boolean function as a product of maxterms.

CANONICAL FORMS

- To convert from one canonical form to another, interchange the symbols ∑ and ∏ and list those numbers missing from the original form.

Truth Table for $F = xy + x'z$

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1



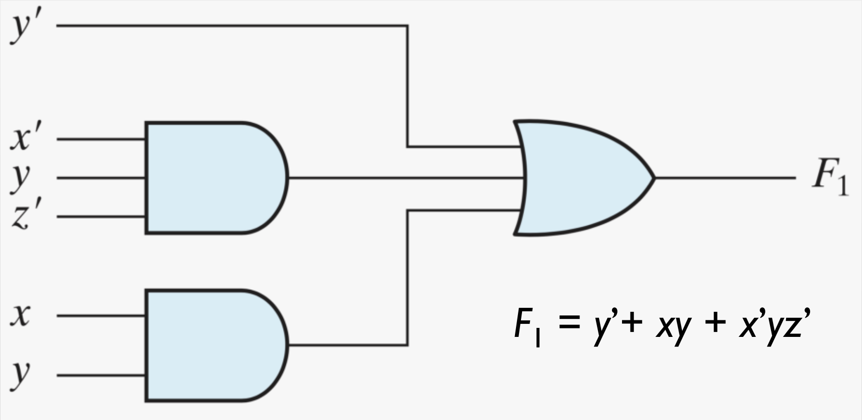
$$F(x, y, z) = \Sigma(1, 3, 6, 7)$$

$$F = xy + x'z$$

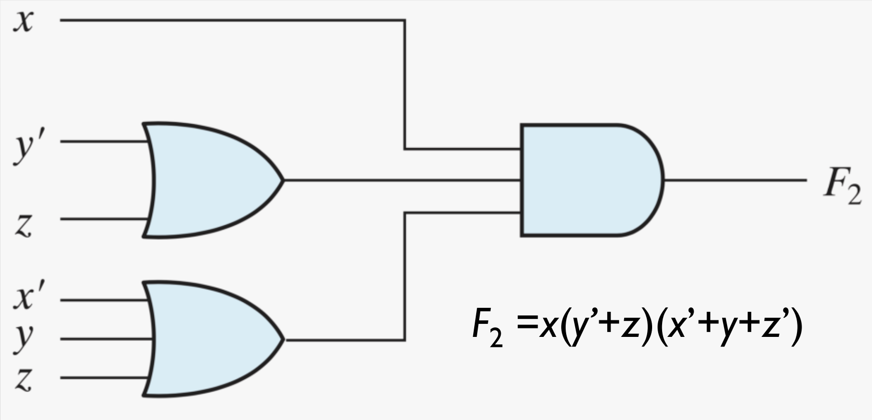
$$F(x, y, z) = \Pi(0, 2, 4, 5)$$

STANDARD FORMS

- In this configuration, the terms that form the function may contain one, two, or any number of literals.
- There are two types of standard forms: the sum of products and product of sums.



(a) Sum of Products



(b) Product of Sums

NON-STANDARD FORMS

- A Boolean function may be expressed in a nonstandard form i.e. neither SOP nor POS.